



## **Banks and Gesso, LLC**

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**MAY 29 2003**

**Division of Minerals and Geology**

Erica S. Crosby  
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Division of Minerals and Geology  
Department of Natural Resources  
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May 28, 2003

TR-2

RE: CEMEX, Inc Lyons Mine Permit No. M-1977-208  
Technical Revision Narrative

CEMEX, Inc. requests a Technical Revision to their Mine Permit No. M-1977-208 to allow an on-site disposal area to dispose of concrete rubble for use at the close and demolition of the concrete plant. The following narrative describes the disposal area. Guidance for this TR was provided by the Division in a letter dated April 7, 2003 indicating the information that was to be included in this narrative. That letter is included with this submittal for reference.

### **Narrative**

This narrative accompanies a request for a Technical Revision (TR) to M-1977-208 (Cemex, Inc. Lyons Cement Plant). The TR adds to the permit a proposed on-site disposal area for the disposal of concrete rubble after the demolition of the concrete plant. The attached On-Site Disposal Cell Map indicates the current plant layout, existing topography, and proposed disposal cell area.

### **Disposal Site**

The location of the disposal site is shown on the On-Site Disposal Cell Map. The site is located south and west of the existing plant site. The footprint encompasses approximately 10 acres. The material will be placed in a wedge shape against the existing hogback as shown by the cross section on the On-Site Disposal Cell Map. The maximum thickness of the material along the hogback will be 45 feet. The disposal area will taper to a thickness of 20 feet. The average thickness of debris placement will be approximately 30 feet.

The liner will be either a geosynthetic liner or a three-foot thick clay liner meeting the permeability requirements of  $1 \times 10^{-6}$  cm/sec. A 3-foot thick layer of cover material will be placed over the liner to protect the liner during the debris placement process. Cross sections on the attached map provide additional information.

#### Groundwater and Surface Water

Groundwater in the area is shallow, approximately 3 feet below ground surface. Only minor regrading will be performed for the placement of the liner material, so neither the liner nor the disposed rubble will be in contact with the water table. Surface drainage will be routed around the disposal area as shown on the attached map. The channels that will route stormwater around the disposal cell will be installed prior to construction of the disposal area and will remain in place after final revegetation. As such, the channels serve as the temporary and permanent diversion structures for the disposal area.

#### Description of Debris

The material to be disposed of on-site will be concrete. The volume of concrete to be disposed is approximately 80,000 cubic yards. This includes the rubble from demolished buildings, silos, foundations, and concrete pavement. Non-concrete material will be removed to an off-site facility.

#### Description of Debris Disposal

A crushing operation will be brought on site during the razing process and the concrete structures and pavement will be crushed to 24-inch or smaller sized blocks. Steel reinforcement will be removed and transported to an off-site facility. The concrete rubble will be transported to the disposal site. Fill dirt will be added to each 3-foot debris lift to fill the voids between the blocks and stabilize the rubble. A 3-foot cap of earth will be placed over the disposal area after final concrete disposal occurs. The disposal area will be revegetated.

#### Hazardous Materials

No hazardous materials will be stored at the plant site at the time of closure. Should any hazardous materials be encountered during the razing of the plant, the materials will be removed off-site and disposed of according to disposal regulations for the material.

We trust that the above narrative has provided the Division with sufficient information regarding the on-site disposal area. Please call if you have any questions regarding this submittal.

Regards,  
Banks and Gesso, LLC



M Danelle Volpe, P.E.

enc: On-Site Disposal Cell Map  
April 7, 2003 letter from the Division

cc: Steve Mossberg, CEMEX, Inc.

# STATE OF COLORADO

## DIVISION OF MINERALS AND GEOLOGY

Department of Natural Resources

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BANKS & GESSO, LLC

Bill Owens  
Governor

Greg E. Walcher  
Executive Director

Ronald W. Cattany  
Division Director

April 7, 2003

Steve Mossberg  
CEMEX Inc.  
P.O. Box 529  
Lyons, CO 80540

**RE:**            *Reclamation Cost Estimate; Disposal of Concrete,  
Lyons Quarry, Permit No. M-1977-208*

Dear Mr. Mossberg,

The Division recently spoke with Danelle Volpe of Banks and Gesso regarding the reclamation cost estimate for the Lyons Quarry, Permit No. M-1977-208. The conversation discussed reclamation costs and requirements for the disposal of concrete on site. The letter sent to you on April 2, 2003 addressed disposal for all structures at the CEMEX site (including concrete, steel, etc.). If CEMEX proposes to only dispose of concrete on site, then the Division would not require extensive testing of material or lining of the pit. The cost associated with the landfilling of concrete is enormous, especially for a site of this magnitude. If CEMEX only proposes to backfill concrete on site, the following criteria will need to be met:

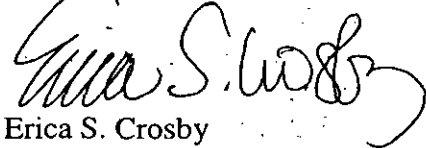
1. A technical revision to the permit must be submitted within 60-days of posting of the financial warranty.
2. A description of the disposal site(s) and their proximity and relationship to ground and surface water. A description of any ground and surface water protection measure to be installed such as landfill liners, covers, diversion ditches, or temporary storm water controls (*if needed*).
3. A description of the debris to be disposed on site, and a description of the types of material to be hauled to a landfill.
4. Provide an estimate of the volume of debris to be landfilled.
5. Provide a brief discussion of the hazardous materials that may be encountered during plant demolition and how they will be managed and disposed.
6. Provide a plan describing how debris will be placed and stabilized in disposal cells to prevent voids and settlement.

7. Locate the disposal sites on a map and provide maps, cross sections, and descriptions of the final configuration of disposal cells. The final graded surfaces of the debris must be free draining.
8. Provide a description of the cover soil that will cap the debris and a map and/or description of the source areas for the cover soil.

The Division hopes that this revised criteria list assists CEMEX in the submittal of the technical revision for the disposal of concrete-only material on site. In addition, the Division will grant an extension to the submittal of reclamation cost estimate to April 21, 2003.

If you have any questions, please feel free to call me at 303-866-4943.

Sincerely,



Erica S. Crosby  
Environmental Protection Specialist

cc: Carl Mount, DMG (*via email*)  
M. Danelle Volpe; Banks & Gesso LLC (*via fax*)